
VALUE-CONDITIONED INTRINSIC SCALING (RND + VCSE)

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ABSTRACT

Exploration remains a core challenge in reinforcement learning, especially in sparse-reward settings. Random Network Distillation (RND) addresses this by rewarding novelty through predictor error, but is value-agnostic and, therefore, may encourage exploration of novel yet task-irrelevant states. Motivated by Value-Conditional State Entropy (VCSE), we propose Value-Conditioned Random Network Distillation (VC-RND), which scales normalized intrinsic rewards using value estimates from a running value mean of state values. We study two scaling variants, inverse linear scaling and sigmoid gating, and evaluate them against PPO, PPO+RND, and PPO+RND+Replay on three DeepMind Control Suite tasks: Cartpole Swingup Sparse, Walker Stand, and Cheetah Run, as well as Two MiniGrid tasks: Multiroom and FourRoom. We analyze mean episodic return, intrinsic reward over time, explained variance, and intrinsic reward distributions. Across DMC tasks, the effect of value-conditioning depends strongly on reward structure. In Cartpole Swingup Sparse, VC-RND performs similarly to other curiosity-based methods, suggesting that value-conditioning has limited effect when extrinsic reward is nearly absent. In Walker Stand, VC-RND substantially improves over PPO and vanilla RND and approaches replay-based performance, indicating that value-aware scaling is most helpful in environments with meaningful but structured reward variation. In Cheetah Run, VC-RND achieves performance comparable to replay-based RND and well above vanilla RND, while also producing a smoother and faster decay in intrinsic reward, suggesting more controlled exploration. In the MiniGrid tests, VC-RND, has mixed performance but still resulted in more controlled exploration. Overall, our results show that value-conditioning can improve exploration, but its effectiveness depends heavily on the structure of the environment.

1 INTRODUCTION

1.1 MOTIVATION

Exploration remains one of the core challenges in reinforcement learning, particularly in sparse-reward environments. Random Network Distillation (RND)(1) provides a scalable and effective intrinsic reward mechanism by defining novelty as the prediction error of a neural network trained to mimic a fixed random target network. This method has enabled substantial progress on hard exploration tasks in sparse-reward environments, such as the Atari game Montezuma’s Revenge. However, RND defines novelty independently of task value. Thus, all novel states are rewarded equally, regardless of whether they are promising for task completion. In supervised RL settings where extrinsic rewards are present, this can lead to the pointless and excessive over-exploration of low-value novel regions.

Recent work on Value-Conditional State Entropy (VCSE)(2) show that standard entropy-based exploration can become biased toward low value states in supervised RL. Because agents frequently visit high-value regions, those states become dense in the visitation distribution, while low-value states remain sparse. Since entropy bonuses reward sparsity, this unintentionally pushes exploration toward low-value but under-visited regions. VCSE fixes this by conditioning entropy estimates on value levels, ensuring that low-value states do not distort exploration around high-value ones.

1.2 VALUE-CONDITIONED RND

Standard RND maximizes state novelty without considering task value. We propose a simple yet principled modification to RND inspired by VCSE, that makes intrinsic reward value-conditioned. Our goal is to introduce a value-aware shaping mechanism for RND that preserves novelty detection, reduces over-exploration of clearly low-value regions, and improves sample efficiency in sparse-reward tasks. We introduce Value-Conditioned Random Network Distillation (VC-RND), a modification to RND that scales intrinsic reward based on state value.

1.3 OBJECTIVES

Our objective with VC-RND being to align intrinsic reward with extrinsic task goals, reduce exploration of irrelevant or low-value states and Improve exploration efficiency. We will achieve this goal by analyzing how VC-RND behaves in different environments and identify the types of environments in which value-conditioning helps. To that end, we evaluate whether VC-RND improves Mean Episodic return, Intrinsic Reward over time, Explain Variance and Intrinsic reward distribution.

2 RELATED WORKS

2.1 PREDICTION-ERROR CURIOSITY METHODS

Random Network Distillation (RND) (1) defines novelty as the prediction error of a predictor network trained to match a fixed randomly initialized target network, and has demonstrated strong performance on tasks such as Montezuma’s Revenge. However, RND remains value-agnostic, rewarding novelty without considering whether a state contributes to task progress. The Intrinsic Curiosity Module (ICM) (3) similarly defines curiosity through prediction error without incorporating task value. Together, these methods establish prediction-error curiosity as a general exploration paradigm, but do not address how novelty should be weighted relative to task relevance.

2.2 STATE-DIFFERENCE AND COUNT-BASED EXPLORATION

Several methods define intrinsic reward through state change or visitation frequency. RIDE (4) rewards actions that produce large changes in a learned state representation, while NovelD (5) uses the difference in RND novelty between consecutive states to encourage boundary-crossing exploration. EME (6) introduces a metric-based bonus with a scaling factor from reward model ensemble variance. On the count-based side, Bellemare et al. (7) generalized count-based exploration to high-dimensional spaces using pseudo-counts, and Tang et al. (8) showed that hash-based approximations can be surprisingly effective. None of these methods incorporate value information when allocating bonuses.

2.3 VALUE-AWARE EXPLORATION

Value-Conditional State Entropy (VCSE) (2) shows that entropy-based exploration can become biased toward low-value states, and fixes this by conditioning entropy on value levels. Although VCSE operates on entropy rather than prediction error, its central insight that exploration should account for value directly motivates our modification of RND. Constrained Intrinsic Motivation (CIM) (9) addresses intrinsic reward bias through constrained policy optimization; our approach instead shapes the reward signal itself through value-based scaling.

2.4 STABILITY AND REPLAY IN INTRINSIC MOTIVATION

Sovrano (2019) (10) extends PPO+RND with prioritized experience replay, noting that intrinsic rewards may interact with replay in non-trivial ways. Analyses of RND (11) further highlight limitations such as catastrophic forgetting, reward scaling sensitivity, and noise vulnerability. Rather than modifying replay or predictor architecture, our approach adjusts how intrinsic rewards influence policy updates through value-based scaling.

2.5 INTRINSIC REWARD SHAPING

AIRS (12) proposes automatic shaping of intrinsic signals, EXPLORS (13) introduces reward-shaping for sparse-reward environments, and DEIR (14) decouples action-relevant novelty from environment stochasticity. Our method can be viewed as a lightweight shaping mechanism that modulates prediction-error intrinsic rewards using critic value estimates, integrating task relevance directly into curiosity-based exploration.

3 METHODOLOGIES

3.1 SETUP

We consider a standard Markov Decision Process (MDP), (S, A, P, R, γ) , with policy $\pi_\theta(a|s)$ and value function $V(s)$. Denote e_t as the extrinsic reward at time t .

3.2 BASELINE: PPO + RND

Baseline RND provides an intrinsic motivation mechanism that augments the environment reward to

$$r_t = e_t + i_t, \quad (1)$$

where i_t is a novelty-based exploration bonus. RND works by defining a fixed, randomly initialized target network f and a predictor network $\hat{f}$.

$$f : O \rightarrow R^k \quad (2)$$

$$\hat{f} : O \rightarrow R^k \quad (3)$$

The predictor network is trained on the output of the random network by minimizing the expected mean squared error:

$$\min_{\theta} E[|\hat{f}(x; \theta) - f(x)|^2] \quad (4)$$

The intrinsic reward is defined as the prediction error of the predictor network:

$$i_t = ||\hat{f}(s_{t+1}) - f(s_t + 1)||^2 \quad (5)$$

3.3 PROPOSED METHOD: VALUE-CONDITIONED RND (VC-RND)

We modify the intrinsic reward by normalizing it and scaling it with a value-dependent function.

Reward Normalization: Because prediction error is non-stationary during training, we normalize the intrinsic reward using running statistics. We maintain exponential moving averages, denote $\tilde{r}_t^{RND}$ as the return at time step t :

$$\mu_t = (1 - \alpha)\mu_{t-1} + \alpha\tilde{r}_t^{RND} \quad (6)$$

$$\sigma_t^2 = (1 - \alpha)\sigma_{t-1}^2 + \alpha(\tilde{r}_t^{RND} - \mu_t)^2 \quad (7)$$

$$r_t^{RND} = \frac{\tilde{r}_t^{RND}}{\sigma_t + \epsilon} \quad (8)$$

This ensures the bonus remains on a stable scale throughout training.

Value-Conditioned Intrinsic Scaling: We define a running mean value estimate $\bar{V}_t$ using moving average:

$$\bar{V}_t = (1 - \beta)\bar{V}_{t-1} + \beta V(s_t) \quad (9)$$

Where $V(s_t)$ is the value of state s_t and β is a tuneable parameter.

Define the value deviation ΔV as:

$$\Delta V_t = |V(s_t) - \bar{V}_t| \quad (10)$$

We will test two different Value-conditioned scaling options.

Option 1: Inverse linear scaling:

$$g(V(s_t)) = \frac{1}{1 + \lambda \Delta V_t} \quad (11)$$

Option 2: Sigmoid Gating:

$$g(V(s_t)) = \sigma(-\lambda \Delta V_t) \quad (12)$$

This will result in a final intrinsic reward of:

$$r_t^{VC-RND} = r_t^{RND} \cdot g(V(s_t)) \quad (13)$$

4 EXPERIMENTS

4.1 BASELINE AND EVALUATION METRICS

Our new purposed models will be compared against PPO, PPO + RND, PPO + RND + Replay Buffer.

We evaluate our method on sparse-reward tasks from the DeepMind Control Suite (DMC), including: Cartpole Swingup (sparse variant), Walker Stand and Cheetah Run, As well as two MiniGrid tasks: FourRooms and Multi-Room. These tasks are all structurally different in terms of extrinsic and intrinsic reward, and will provide deep insight on how VC-RND preforms in different environments. We evaluate the algorithms using Mean episodic return, Intrinsic Reward over time, Explain Variance and Intrinsic reward distribution.

4.2 DMC EXPERIMENTAL SETUP

The following experimental setup was used: Parallel environments: 16. Timesteps: 500k. Timesteps Carpole Swingup: 1 million. Intrinsic reward normalization: Applied. Epochs: 10. Mini batches: 8. Seeds: 3.

4.3 MINI EXPERIMENTAL SETUP

The following experimental setup was used: Parallel environments: FourRooms: 64, Multi-Room: 16. Timesteps: FourRooms: 1 million, Multi-Room: 3 million. Intrinsic reward normalization: Applied. Epochs: 4. Mini batches: 4. Seeds: 3.

5 RESULTS

5.1 DMC RESULTS:

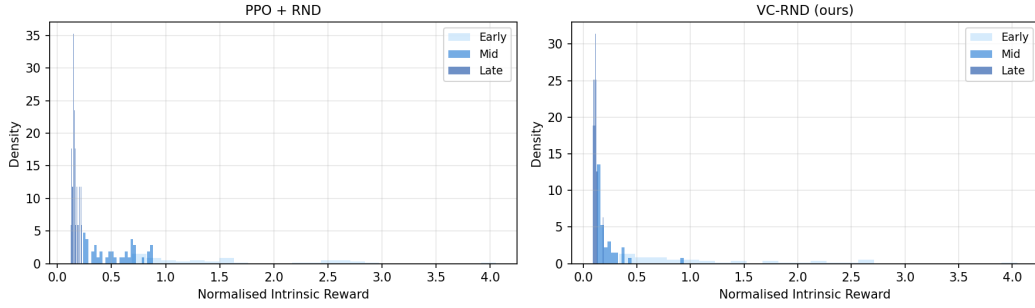
Table 1: Final mean episodic return for DMC environments across algorithms.

Environment	Algorithm	Last	Last-10 Avg
Cartpole	PPO	0.00	0.00
Cartpole	PPO+RND	2.02	2.17
Cartpole	PPO+RND+Replay	1.53	1.73
Cartpole	VC-RND Linear	2.51	2.53
Walker	PPO	274.99	274.41
Walker	PPO+RND	338.57	337.14
Walker	PPO+RND+Replay	369.88	365.38
Walker	VC-RND Linear	366.02	360.06
Walker	VC-RND Sigmoid	339.17	340.28
Cheetah	PPO	182.12	177.46
Cheetah	PPO+RND	49.62	48.99
Cheetah	PPO+RND+Replay	93.21	89.27
Cheetah	VC-RND Linear	93.81	91.94
Cheetah	VC-RND Sigmoid	94.99	92.31

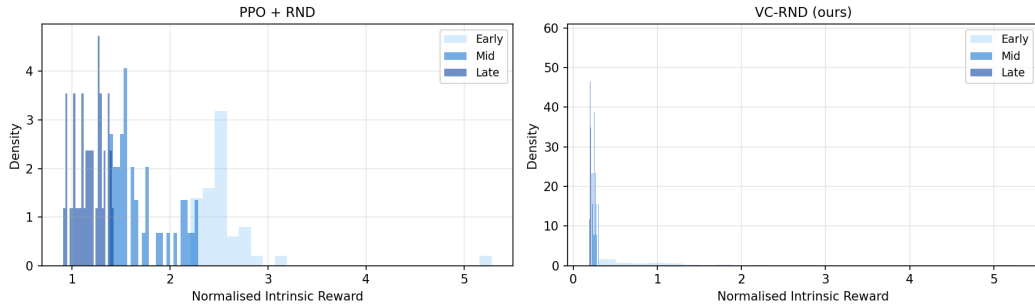
Table 2: Explained variance stability statistics for the DMC environments.

Env.	Algo.	Low	Peak	Diff.
Walker	PPO	-2.63127	0.70741	3.33868
Walker	PPO+RND	-0.26690	0.59561	0.86251
Walker	PPO+RND+Replay	-0.26690	0.52528	0.79218
Walker	VC-RND	-0.26690	0.55289	0.81979
Walker	VC-RND Sigmoid	-0.26690	0.49699	0.76389
Cheetah	PPO	-9.57200	0.94587	10.51787
Cheetah	PPO+RND	-9.57200	0.85151	10.42351
Cheetah	PPO+RND+Replay	-9.57200	0.86785	10.43985
Cheetah	VC-RND	-9.57200	0.90838	10.48038
Cheetah	VC-RND Sigmoid	-9.57200	0.93081	10.50281

Intrinsic Reward Distribution — Cheetah Run



Intrinsic Reward Distribution — Walker Stand



VC-RND Analysis — Cartpole Swingup Sparse

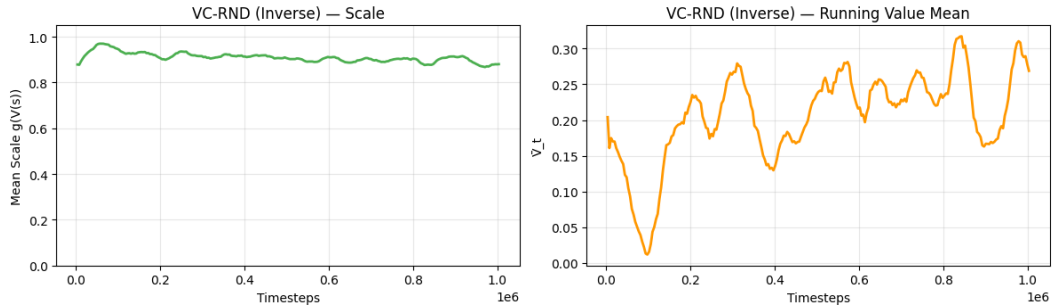


Table 3: Final mean episodic return for MiniGrid FourRooms (1M steps).

Environment	Algorithm	Last	Last-10% Avg
FourRooms	PPO	0.339	0.320 $\pm$ 0.051
FourRooms	PPO+RND	0.056	0.029 $\pm$ 0.015
FourRooms	PPO+RND+Replay	0.018	0.020 $\pm$ 0.006
FourRooms	VC-RND Linear	0.018	0.030 $\pm$ 0.012
FourRooms	VC-RND Sigmoid	0.023	0.036 $\pm$ 0.019

Table 4: Intrinsic reward distribution statistics for MiniGrid MultiRoom. VC-RND produces a sharply collapsed distribution, indicating aggressive suppression of curiosity as the value function becomes informative.

Algorithm	Phase	Mean	Spread (IQR)	Peak Density
PPO+RND	Early	$\sim$ 6.0	$\sim$ 4.0	$\sim$ 0.4
PPO+RND	Mid	$\sim$ 7.5	$\sim$ 3.5	$\sim$ 0.5
PPO+RND	Late	$\sim$ 10.0	$\sim$ 1.5	$\sim$ 1.2
VC-RND	Early	$\sim$ 0.5	$\sim$ 0.8	$\sim$ 15.0
VC-RND	Mid	$\sim$ 0.2	$\sim$ 0.3	$\sim$ 22.0
VC-RND	Late	$\sim$ 0.1	$\sim$ 0.2	$\sim$ 23.0

5.2 MINIGRID RESULTS

6 DISCUSSION

6.1 DMC DISCUSSION

The DMC ablations show that the effect of value-conditioned intrinsic scaling is highly environment-dependent.

In Cartpole Swingup Sparse, all curiosity-based methods performed similarly, with VC-RND Linear achieving the best final (2.51) and average (2.53) mean episodic return, but not by much. PPO without intrinsic reward failed entirely, while all curiosity-based variants reached similarly low but non-zero returns. This suggests that intrinsic motivation is necessary in this task, but value-conditioning offers limited benefit. This makes sense when we consider the structure of the Sparse Cartpole Environment. Rewards are almost always zero, with non-zero rewards only being 1. As a result, value scaling had little effect, causing VC-RND to behave just like RND. The VC-RND analysis provides evidence for this claim as VC scaling stays roughly the same (0.9) throughout. In summary, VC-RND behaves similarly to standard RND because most exploration is still driven by novelty rather than value-aware filtering.

In Walker Stand, VC-RND is much more effective. Both VC-RND variants clearly outperform PPO and vanilla PPO+RND, and the linear version nearly matches PPO+RND+Replay (366.02 vs 369.88), which is the strongest method on both final return and average return. This is the clearest case where value-conditioning helps. Walker provides a dense but structured environment, where rewards were continuous values with a great deal of variation, and therefore controlled exploration was beneficial. Thus, serving as an optimal situation for VC-RND. The Sigmoid variant of VC-RND achieved the smallest difference from low to peak explained variance, indicating it had stable, accurate value function predictions. The benefit of VC-RND, specifically Linear VC-RND is no where better visualized than when looking at the intrinsic reward distribution. RND’s intrinsic reward distribution is far wider spread, compared to the VC-RND distribution. Indicating noisy exploration. VC-RND’s narrower, right-skewed intrinsic distribution indicates more controlled and focused exploration.

In Cheetah Run, PPO strongly outperforms all curiosity-based methods, indicating that this dense-reward environment does not benefit from exploration bonuses. However, within the curiosity-based methods, both VC-RND variants substantially outperform vanilla PPO+RND and achieve performance close to PPO+RND+Replay. This suggests that although intrinsic reward is somewhat harmful in Cheetah overall, value-conditioning reduces that harm by value-conditioning curiosity. The narrower intrinsic-reward distribution indicates that the agent was being mindful of state values,

and thus making smarter exploration choices. VC-RND does not make Cheetah a strong use case for intrinsic motivation, but it does make curiosity less noisy and less disruptive than vanilla RND.

Comparing the two scaling choices, the inverse linear variant is generally the stronger and more consistent option. It gives the best result in Cartpole and Walker, while sigmoid gating is only slightly better in Cheetah. This suggests that a smoother, less aggressively scaling rule may preserve useful curiosity better than the sigmoid version. The ablations indicate that the exact form of scaling matters and may need to be adapted to environment reward structure.

Overall, the DMC ablations suggest a clear pattern. VC-RND is not universally superior, but it is most useful in environments where novelty alone is too broad and where value estimates are informative enough to shape exploration. It brings the largest benefit in structured control tasks such as Walker, where extrinsic reward is present with a great deal of variation. This supports the core hypothesis behind the method, value-conditioning is most beneficial when exploration must be guided, not merely increased.

6.2 MINIGRID DISCUSSION

The MiniGrid results reinforce the same pattern observed in DMC: value-conditioning effectiveness depends on environment structure.

In FourRooms, PPO alone achieves the highest return (0.339), outperforming all curiosity-based methods which remain below 0.04. The VC-RND scaling factor stays near 1.0 ($\bar{g} \approx 0.997$) throughout training, meaning value-conditioning barely activates, just like in Cartpole Sparse, causing VC-RND to behave just like RND. The intrinsic reward distributions confirm this, with PPO+RND and VC-RND producing nearly identical spreads across the 3.5-6.0 range.

In MultiRoom, all methods achieve near-zero return, but the intrinsic reward distributions reveal a striking difference. PPO+RND maintains a wide, rightward-shifting distribution (mean ~ 6.0 early to ~ 10.0 late), while VC-RND collapses to near zero (peak density $\sim 15\text{--}23\times$ higher). The longer horizon and higher intrinsic coefficient (1.0 vs 0.02) produce larger value deviations, giving the scaling function more room to act. However, this aggressive dampening did not translate into task progress, suggesting that suppressing curiosity too strongly in very sparse environments may be counterproductive when the agent has not yet discovered the goal.

A broader issue in MiniGrid is that all RND-based methods, including VC-RND, struggled relative to PPO. This is partly attributable to poor state representations under partial observability. MiniGrid provides small pixel observations from a limited field of view, making it difficult for the RND predictor to distinguish meaningfully between novel and familiar states. As a result, the RND prediction error remains noisy and fails to decay over training, which is visible in the persistently high normalized intrinsic reward (~ 4.0) across all RND variants in FourRooms. Despite this, within the RND-based methods, the value-conditioned approach did produce measurably more controlled exploration, as evidenced by the tighter intrinsic reward distributions in both environments. This suggests that even when the underlying novelty signal is noisy, value-conditioning can still reduce exploration variance, though translating that into improved task performance requires a stronger base representation than MiniGrid’s partial observations currently provide.

7 LIMITATIONS AND FUTURE WORK

7.1 LIMITATIONS

Our study has several important limitations that should be acknowledged. First, all experiments were conducted with only three random seeds, which limits the statistical robustness of our findings. Observed performance differences, particularly in Cartpole Swingup Sparse where margins between methods were small, may reflect seed-level variance rather than consistent algorithmic advantages.

Second, the hyperparameter sensitivity of VC-RND remains underexplored. The scaling parameter λ , the moving average coefficient β for the running value mean, and the intrinsic reward coefficient all interact in ways that may vary across environments. Our MiniGrid experiments highlighted this issue: FourRooms required an intrinsic coefficient of 0.02, while Multi Room required 1.0, suggesting that VC-RND may demand more careful tuning than standard RND. We did not con-

duct a systematic hyperparameter sweep, so the reported results may not reflect the best achievable performance for each variant.

Third, our evaluation was limited to a narrow set of environments from the DeepMind Control Suite and MiniGrid. These are relatively low-dimensional continuous and discrete control tasks. We did not evaluate on high-dimensional observation spaces such as Atari games, where RND was particularly effective. The extent to which VC-RND transfers to such settings remains unknown.

Fourth, our method uses a single global running value mean $\bar{V}_t$ to define the deviation ΔV_t . This assumes a unimodal value distribution across visited states, which may not hold in environments with multiple distinct reward regions. In such settings, a single baseline may fail to distinguish meaningfully between high-value and low-value states, reducing the precision of the scaling function.

Finally, the two scaling variants we tested, inverse linear and sigmoid gating, are both fixed functional forms chosen a priori. Neither adapts to the evolving reward landscape during training. A learned or adaptive scaling function could potentially offer better performance, but would also introduce additional complexity and optimization challenges.

7.2 FUTURE WORK

Several directions could address the limitations above and extend the contributions of this work. The most immediate is to scale experiments to a larger number of seed and a broader set of environments, including Atari games with sparse rewards such as Montezuma’s Revenge and Pitfall. Testing on procedurally generated environments such as ProcGen would also help assess whether value-conditioning generalizes across environment distributions.

An important algorithmic extension would be adaptive scaling. Rather than using a fixed λ and a predetermined functional form for $g(V(s_t))$, a learned scaling function could be trained jointly with the policy.

Combining VC-RND with experience replay mechanisms is another natural extension. Our results showed that PPO+RND+Replay was a strong baseline, particularly in Walker Stand. Integrating value-conditioned scaling with prioritized replay could yield further gains, as replay may help stabilize both the value function and the intrinsic reward signal simultaneously.

Finally, extending the value-conditioning idea beyond RND to other intrinsic motivation methods such as ICM or count-based exploration would help determine whether value-aware scaling is a general principle or one that is specific to prediction-error curiosity. If the benefits transfer broadly, this would suggest that value-conditioning is a useful modular component for any exploration bonus, rather than a method-specific modification.

8 CONCLUSION

We proposed VC-RND, a value-aware extension of Random Network Distillation that scales normalized intrinsic reward using deviation from a running value baseline. The goal was not simply to increase exploration, but to make curiosity more controlled and better aligned with task progress. Across DMC environments, the results show that this idea is promising but environment-dependent. In Cartpole Swingup Sparse, value-conditioning had only limited impact because the sparse reward signal produced weak value estimates. In Walker Stand, VC-RND delivered its strongest results, outperforming PPO and vanilla RND and approaching replay-based RND, indicating that value-aware exploration is particularly effective in structured environments with meaningful reward variation. In Cheetah Run, VC-RND did not surpass PPO, but it improved substantially over vanilla RND and produced smoother intrinsic-reward dynamics, suggesting that value-conditioning can reduce the noise introduced by curiosity in dense-reward settings. In the MiniGrid discrete navigation tasks, VC-RND showed limited return improvement over standard RND, as the sparse binary reward structure produced weak value estimates that kept the scaling factor near 1.0. However, the intrinsic reward distributions in MultiRoom showed that when the value signal is even marginally informative, VC-RND can substantially compress exploration noise, even if this does not yet translate into task progress. Overall, these results support the claim that value-conditioning can improve curiosity-driven exploration, but only when the critic provides a sufficiently informative signal.

9 TEAM CONTRIBUTIONS

Pranshu Sharma conducted the literature review, Limitations and Future Work section, and conducted the MiniGrid test as well as their analysis. Hassan Ghouri completed the Abstraction, Introduction, Methodologies, and conducted the DMC test as well as their analysis.

REFERENCES

- [1] Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by random network distillation. *arXiv preprint arXiv:1810.12894*, 2018.
- [2] Dongyoung Kim, Jinwoo Shin, Pieter Abbeel, and Younggyo Seo. Accelerating reinforcement learning with value-conditional state entropy exploration. *arXiv preprint arXiv:2305.19476*, 2023.
- [3] Deepak Pathak, Pulkit Agrawal, Alexei A. Efros, and Trevor Darrell. Curiosity-driven exploration by self-supervised prediction. *arXiv preprint arXiv:1705.05363*, 2017.
- [4] Roberta Raileanu and Tim Rocktäschel. Ride: Rewarding impact-driven exploration for procedurally-generated environments. In *International Conference on Learning Representations*, 2020.
- [5] Tianjun Zhang, Huazhe Xu, Xiaolong Wang, Yi Wu, Trevor Darrell, Joseph E Gonzalez, and Vibhav Vineet. Noveld: A simple yet effective exploration criterion. In *Advances in Neural Information Processing Systems*, 2021.
- [6] Yiming Wang, Kaiyan Zhao, Furui Liu, and Leong Hou U. Rethinking exploration in reinforcement learning with effective metric-based exploration bonus. In *Advances in Neural Information Processing Systems*, 2024.
- [7] Marc G. Bellemare, Sriram Srinivasan, Georg Ostrovski, Tom Schaul, David Saxton, and Rémi Munos. Unifying count-based exploration and intrinsic motivation. In *Advances in Neural Information Processing Systems*, 2016.
- [8] Haoran Tang, Rein Houthoofd, Davis Foote, Adam Stooke, Xi Chen, Yan Duan, John Schulman, Filip De Turck, and Pieter Abbeel. #exploration: A study of count-based exploration for deep reinforcement learning. In *Advances in Neural Information Processing Systems*, 2017.
- [9] Xiang Zhao, Rui Xu, Songbo Li, Shuai Liu, Siyuan Li, Chenjia Bai, Zhen Wang, and Xuelong Li. Constrained intrinsic motivation for reinforcement learning. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, 2024.
- [10] Francesco Sovrano. Combining experience replay with exploration by random network distillation. *arXiv preprint arXiv:1905.07579*, 2019.
- [11] Hassan Ghouri. Csc415 a1: Review and analysis of random network distillation. Technical report, University of Toronto Mississauga, 2026. Course assignment report.
- [12] Mingqi Yuan, Bo Li, Xin Jin, and Wenjun Zeng. Automatic intrinsic reward shaping for exploration in deep reinforcement learning. *arXiv preprint arXiv:2301.10886*, 2023.
- [13] Rati Devidze, Parameswaran Kamalaruban, and Adish Singla. Exploration-guided reward shaping for reinforcement learning under sparse rewards. In *Advances in Neural Information Processing Systems*, 2022.
- [14] Wenqi Hao, Jiachen Hao, Yitong Liu, and Yifan Chen. Deir: Efficient and robust exploration through discriminative-model-based episodic intrinsic rewards. *arXiv preprint arXiv:2304.10770*.